RAJALAKSHMI ENGINEERING COLLEGE

Monika S 251801189-2025-AIDS

M2

Started on	Wednesday, 30 September 2026, 11:19 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 11:19 AM
Time taken	33 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int duplicate = -1;

    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[i] == arr[j]) {
                duplicate = arr[i];
                break;
            }
        }
        if (duplicate != -1) {
            break;
        }
    }

    printf("%d\n", duplicate);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔

	Input	Expected	Got	
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Monika S 251801189-2025-AIDS

M2

Started on	Wednesday, 30 September 2026, 11:20 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 11:21 AM
Time taken	1 min 25 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find Duplicate in Array.

Given a read only array of n integers between 1 and n, find one number that repeats.

Input Format:

First Line - Number of elements

n Lines - n Elements

Output Format:

Element x - That is repeated

For example:

Input	Result
5 1 1 2 3 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int slow = arr[0];
    int fast = arr[0];

    do {
        slow = arr[slow];
        fast = arr[arr[fast]];
    } while (slow != fast);

    slow = arr[0];
    while (slow != fast) {
        slow = arr[slow];
        fast = arr[fast];
    }

    printf("%d\n", slow);

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	11 10 9 7 6 5 1 2 3 8 4 7	7	7	✔
✔	5 1 2 3 4 4	4	4	✔

	Input	Expected	Got	
✔	5 1 1 2 3 4	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Monika S 251801189-2025-AIDS

M2

Started on	Wednesday, 30 September 2026, 11:21 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 11:22 AM
Time taken	45 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Find the intersection of two sorted arrays.

OR in other words,

Given 2 sorted arrays, find all the elements which occur in both the arrays.

Input Format

· The first line contains T, the number of test cases. Following T lines contain:

1. Line 1 contains N1, followed by N1 integers of the first array

2. Line 2 contains N2, followed by N2 integers of the second array

Output Format

The intersection of the arrays in a single line

Example

Input:

1

3 10 17 57

6 2 7 10 15 57 246

Output:

10 57

Input:

1

6 1 2 3 4 5 6

2 1 6

Output:

1 6

For example:

Input	Result
1 3 10 17 57 6 2 7 10 15 57 246	10 57

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

void solve() {
    int n1, n2;

    // Read first array size and its elements
    if (scanf("%d", &n1) != 1) return;
    int *arr1 = (int *)malloc(n1 * sizeof(int));
    for (int i = 0; i < n1; i++) {
        scanf("%d", &arr1[i]);
    }

    // Read second array size and its elements
```

```c
    if (scanf("%d", &n2) != 1) {
        free(arr1);
        return;
    }
    int *arr2 = (int *)malloc(n2 * sizeof(int));
    for (int i = 0; i < n2; i++) {
        scanf("%d", &arr2[i]);
    }

    int i = 0, j = 0;
    int first = 1;

    // Two-pointer intersection logic
    while (i < n1 && j < n2) {
        // Skip duplicates in arr1
        if (i > 0 && arr1[i] == arr1[i - 1]) {
            i++;
            continue;
        }

        if (arr1[i] < arr2[j]) {
            i++;
        } else if (arr2[j] < arr1[i]) {
            j++;
        } else {
            if (!first) {
                printf(" ");
            }
            printf("%d", arr1[i]);
            first = 0;
            i++;
            j++;
        }
    }
    printf("\n");

    free(arr1);
    free(arr2);
```

	Input	Expected	Got	
✔	1 3 10 17 57 6 2 7 10 15 57 246	10 57	10 57	✔
✔	1 6 1 2 3 4 5 6 2 1 6	1 6	1 6	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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Monika S 251801189-2025-AIDS

M2

Started on	Wednesday, 30 September 2026, 11:23 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 11:23 AM
Time taken	42 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```
#include <stdio.h>
#include <stdlib.h>

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) {
        free(arr);
        return 0;
    }

    int found = 0;


    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (arr[j] - arr[i] == k || arr[i] - arr[j] == k) {
                found = 1;
                break;
            }
        }
        if (found) break;
    }

    printf("%d\n", found);
```

```
33
34      free(arr);
35      return 0;
36  }
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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RAJALAKSHMI ENGINEERING COLLEGE

Monika S 251801189-2025-AIDS

M2

Started on	Wednesday, 30 September 2026, 11:23 AM
State	Finished
Completed on	Wednesday, 30 September 2026, 11:26 AM
Time taken	2 mins 17 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.

Input Format:

First Line n - Number of elements in an array

Next n Lines - N elements in the array

k - Non - Negative Integer

Output Format:

1 - If pair exists

0 - If no pair exists

Explanation for the given Sample Testcase:

YES as 5 - 1 = 4

So Return 1.

For example:

Input	Result
3 1 3 5 4	1

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

int findPairWithDifference(int *arr, int n, int k) {
    int i = 0;
    int j = 1;

    // Two-pointer approach on a sorted array
    while (i < n && j < n) {
        if (i != j && arr[j] - arr[i] == k) {
            return 1;
        } else if (arr[j] - arr[i] < k) {
            j++;
        } else {
            i++;
        }
    }
    return 0;
}

int main() {
    int n;
    if (scanf("%d", &n) != 1) return 0;

    int *arr = (int *)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int k;
    if (scanf("%d", &k) != 1) {
        free(arr);
```

```
        return 0;
    }

    printf("%d\n", findPairWithDifference(arr, n, k));

    free(arr);
    return 0;
}
```

	Input	Expected	Got	
✔	3 1 3 5 4	1	1	✔
✔	10 1 4 6 8 12 14 15 20 21 25 1	1	1	✔
✔	10 1 2 3 5 11 14 16 24 28 29 0	0	0	✔
✔	10 0 2 3 7 13 14 15 20 24 25 10	1	1	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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